# UNIT IV

BI – DATA ANALYTICS PROCESS - Introduction to analytics process, Types of Analytical Techniques in BI –Descriptive, Predictive, Perspective, Social Media Analytics, Behavioral, Iris Datasets.

**CO 4** Understand & Implement the K-Means Clustering with Iris Dataset- K2, K3

# Lecture 25

**BI – Data Analytics Process**

**What is Analytics?**

**Analytics** is the **systematic process of examining data** to discover meaningful insights, detect patterns, and support **decision-making**.

In BI, analytics turns **raw data** into **useful information** that helps organizations:

- Improve efficiency
- Understand customers
- Predict future trends
- Gain competitive advantage

**What is the Analytics Process?**

The **Analytics Process** is a **step-by-step approach** that guides how raw data is transformed into actionable insights.

It ensures that analysis is:

- **Systematic** (organized)
- **Repeatable** (can be applied again)
- **Actionable** (useful for decisions)

**Stages of the Analytics Process**

1. **Define the Problem / Business Objective**
    - Clearly identify the **goal** (e.g., “Why are sales declining in region X?”).
2. **Data Collection**
    - Gather relevant data from multiple sources: databases, warehouses, IoT devices, CRM, ERP, social media, etc.
3. **Data Preparation (Preprocessing)**
    - Cleaning (remove errors, duplicates)
    - Integration (combine from multiple sources)

    - Transformation (normalize, standardize, discretize)
    - Reduction (sampling, PCA, feature selection)
4. **Data Exploration (Descriptive Analytics)**
    - Summarize data (mean, median, trends).
    - Use visualization tools (charts, dashboards, OLAP).
5. **Modeling & Analysis**
    - Apply advanced methods like:
        - **Predictive models** (regression, decision trees, ML)
        - **Classification & clustering**
        - **Statistical analysis**
    - Identify hidden patterns and relationships.
6. **Interpretation of Results**
    - Translate models and statistics into **business insights**.
    - Example: "Customers aged 18–25 are more likely to respond to discounts."
7. **Communication & Reporting**
    - Present results through **dashboards, reports, visualizations**.
    - Must be clear, actionable, and tailored to decision-makers.
8. **Decision & Action**
    - Use insights to make **data-driven decisions**.
    - Example: Launch targeted marketing campaigns.
9. **Evaluation & Feedback**
    - Assess whether decisions improved performance.
    - Refine the analytics process for future use.

**Types of Analytics in BI**

1. **Descriptive Analytics** – *What happened?*
    - Example: Monthly sales reports, dashboards.
2. **Diagnostic Analytics** – *Why did it happen?*
    - Example: Analyzing causes of sales drop.
3. **Predictive Analytics** – *What is likely to happen?*
    - Example: Forecasting next quarter's sales.
4. **Prescriptive Analytics** – *What should we do?*
    - Example: Recommending best pricing strategies.

**Importance of the Analytics Process in BI**

- Converts **data into knowledge**.
- Supports **evidence-based decisions**.
- Helps organizations stay **competitive**.
- Identifies **trends, risks, and opportunities**.
- Improves **efficiency and profitability**.

**In short:**
The **Analytics Process** in BI is a structured approach that starts with a **business question**, moves through **data collection, cleaning, and analysis**, and ends with **actionable insights** that drive smart decision-making.

# Lecture 26

## Types of Analytical Techniques in BI

In **Business Intelligence (BI)**, different analytical techniques are used depending on **what type of insight** is required. These techniques help organizations move from **raw data → insights → decision-making**.

### 1. Descriptive Analytics – *"What happened?"*

- Focus: Summarizes **past data** into meaningful information.
- Tools: Dashboards, reports, OLAP, data visualization.
- Example:
  - Monthly sales reports
  - Customer churn statistics
  - Profit and loss statements

Helps in **understanding history and trends**.

### 2. Diagnostic Analytics – *"Why did it happen?"*

- Focus: Identifies **causes** behind past outcomes.
- Techniques: Drill-down analysis, correlation analysis, root cause analysis.
- Example:
  - Why sales dropped in Q3?
  - Why website traffic decreased?

Helps in **understanding reasons and patterns**.

### 3. Predictive Analytics – *"What is likely to happen?"*

- Focus: Uses **statistical models & machine learning** to forecast future outcomes.
- Techniques: Regression, classification, clustering, time series analysis.
- Example:
  - Predicting customer churn
  - Forecasting demand for next quarter
  - Credit risk scoring

Helps in **anticipating future events**.

**4. Prescriptive Analytics – *"What should we do?"***

- Focus: Suggests **actions or strategies** to achieve desired results.
- Techniques: Optimization, simulation, decision trees, recommendation systems.
- Example:
  - Best pricing strategy for maximizing profit
  - Optimizing supply chain routes
  - Personalized product recommendations

Helps in **decision-making and strategy**.

**5. Cognitive Analytics – *"What is the best action using AI?"***

- Focus: Mimics **human reasoning** using **AI, NLP, and deep learning**.
- Techniques: Machine learning, natural language processing, AI-based decision-making.
- Example:
  - Chatbots answering customer queries
  - Fraud detection systems
  - AI-driven medical diagnosis

Helps in **automating intelligent decision-making**.

**Summary Table**

| Type of Analytics | Key Question | Purpose | Example |
|---|---|---|---|
| **Descriptive** | What happened? | Summarize past data | Sales reports |
| **Diagnostic** | Why did it happen? | Find causes & correlations | Drop in revenue |
| **Predictive** | What will happen? | Forecast future outcomes | Customer churn prediction |
| **Prescriptive** | What should we do? | Suggest actions | Supply chain optimization |
| **Cognitive (AI)** | How to act intelligently? | AI-based reasoning | Chatbots, fraud detection |

**In short:**
The main analytical techniques in BI are **Descriptive, Diagnostic, Predictive, Prescriptive,**

**and Cognitive Analytics**. Together, they allow businesses to move from **knowing the past → understanding the cause → predicting the future → deciding the best action**.

# Lecture 27

## Descriptive Analytics

### What is Descriptive Analytics?

**Descriptive Analytics** is the process of analyzing **historical data** to answer the question:

**"What happened?"**

It focuses on **summarizing past events** and presenting them in a meaningful way, usually through **reports, dashboards, and visualizations**.

### Purpose of Descriptive Analytics

- Provide a **clear picture of past performance**
- Identify **trends and patterns** over time
- Help organizations **monitor business activities**
- Act as a **foundation** for advanced analytics (diagnostic, predictive, prescriptive)

### Techniques Used in Descriptive Analytics

1. **Data Aggregation** – Collect and summarize data (totals, averages, counts).
2. **Data Visualization** – Charts, graphs, dashboards, heatmaps.
3. **OLAP (Online Analytical Processing)** – Drill-down, roll-up, slice and dice of data cubes.
4. **Statistical Summaries** – Mean, median, standard deviation, frequency distributions.
5. **Reporting** – Static or dynamic reports (daily sales, monthly revenue).

### Examples of Descriptive Analytics

- A retail chain analyzing **monthly sales reports**.
- A hospital reviewing **patient admission trends** in the last year.
- A bank generating a **report of credit card transactions**.
- A university tracking **student enrollment growth**.
- A company using **dashboards to monitor KPIs** like revenue, expenses, churn.

**Advantages**

- Easy to **understand and interpret**.
- Provides a **baseline** for business performance.
- Supports **monitoring and reporting**.
- Helps spot **general patterns and anomalies**.

**Limitations**

- Only tells **what happened**, not **why**.
- Cannot predict future events.
- Provides **historical insights** but not prescriptive solutions.

**Real-World Use Case**

Example: An **e-commerce company** uses descriptive analytics to:

- Generate **weekly sales reports**
- Track **top-selling products**
- Monitor **website traffic trends**

This helps them understand **past performance**, but to know **why sales increased/decreased**, they need **diagnostic analytics**.

**In short:**
**Descriptive Analytics = Summarizing Past Data.**
It is the **first step** in BI analytics, helping businesses understand **what has already happened**, usually through reports, dashboards, and visual summaries.

# Lecture 28

**Predictive Analytics**

**What is Predictive Analytics?**

**Predictive Analytics** uses **statistical models, machine learning, and data mining techniques** on historical data to answer the question:

**"What is likely to happen in the future?"**

It identifies **patterns and trends** in past data to make **forecasts and predictions**.

**Purpose of Predictive Analytics**

- Anticipate **future trends and behaviors**
- Reduce **risks and uncertainties**
- Improve **decision-making** with data-driven forecasts
- Enable **proactive actions** instead of reactive responses

**Techniques Used in Predictive Analytics**

1. **Regression Analysis**
    - Predicts numeric values.
    - Example: Predicting next month's sales.
2. **Classification**
    - Predicts categorical outcomes.
    - Example: Will a customer churn? (Yes/No).
3. **Clustering**
    - Groups similar data for future trend prediction.
    - Example: Grouping customers into segments for targeted marketing.
4. **Time Series Forecasting**
    - Predicts values based on time-dependent data.
    - Example: Stock price prediction, demand forecasting.
5. **Machine Learning Models**
    - Decision Trees, Random Forest, Neural Networks, SVMs.

**Examples of Predictive Analytics**

- **Retail** → Forecasting demand for the next season.
- **Banking** → Predicting loan default risk.
- **Healthcare** → Predicting patient readmission chances.
- **E-commerce** → Recommending products to customers.
- **Telecom** → Predicting customer churn (who is likely to leave).
- **Manufacturing** → Predicting machine failures (predictive maintenance).

**Advantages**

- Helps businesses **plan for the future**.
- Improves **customer targeting**.
- Reduces **risks** by anticipating problems.
- Supports **better resource allocation**.

**Limitations**

- Predictions are **probabilistic, not certain**.
- Requires **large amounts of quality data**.
- Models can be **complex and expensive** to implement.
- Sensitive to **changes in data patterns** (concept drift).

**Real-World Use Case**

Example: **Netflix** uses predictive analytics to:

- Analyze past viewing history
- Recommend shows and movies users are likely to watch
- Forecast demand for new content

This increases **customer engagement and retention**.

**In short:**
**Predictive Analytics = Forecasting the Future.**
It applies **statistical models & machine learning** to past data in order to **anticipate upcoming events, risks, or opportunities**.

# Lecture 29

**Prescriptive Analytics**

**What is Prescriptive Analytics?**

**Prescriptive Analytics** goes **beyond predicting the future** (predictive) and answers:

**"What should we do to achieve the best outcome?"**

It recommends **actions or strategies** using optimization, simulation, and advanced algorithms.

**Purpose of Prescriptive Analytics**

- Suggest the **best course of action**
- Optimize **resources, processes, and strategies**
- Provide **decision support** for complex problems
- Help organizations **achieve desired goals** efficiently

**Techniques Used in Prescriptive Analytics**

1. **Optimization Models**
    - Finds the best solution within constraints.
    - Example: Minimize delivery costs by optimizing routes.
2. **Simulation**

    - Tests different scenarios before taking action.
    - Example: Simulating demand under different pricing strategies.
3. **Decision Trees & Rule-based Systems**
    - Helps compare alternative decisions.
4. **Machine Learning + Optimization**
    - Learns from past data and recommends the best decision.
5. **What-if Analysis**
    - Evaluates outcomes of different choices.
    - Example: "What if we increase the budget by 10%?"

**Examples of Prescriptive Analytics**

- **Retail** → Suggesting best discount levels to maximize profit.
- **Healthcare** → Recommending personalized treatment plans.
- **Logistics** → Optimizing delivery routes to reduce fuel costs.
- **Finance** → Suggesting best investment portfolios.
- **E-commerce** → Recommending cross-selling and upselling strategies.

**Advantages**

- Provides **clear recommendations** for decision-making.
- Helps achieve **optimal outcomes**.
- Turns insights into **actionable strategies**.
- Reduces risks by simulating different scenarios.

**Limitations**

- Requires **complex models and advanced tools**.
- Dependent on **accurate predictive models**.
- Can be **computationally expensive**.
- Not 100% accurate — real-world factors may differ.

**Real-World Use Case**

Example: **Uber** uses prescriptive analytics for:

- Determining **dynamic pricing (surge pricing)**
- Matching **drivers to passengers** efficiently
- Optimizing **routes** for minimal wait times

This ensures **maximum customer satisfaction and revenue**.

**In short:**
**Prescriptive Analytics = Action-Oriented Analytics.**
It doesn't just tell you **what will happen**; it tells you **what you should do about it** to achieve the best results.

# Lecture 30

## Social Media Analytics (SMA)

### What is Social Media Analytics?

**Social Media Analytics** is the process of collecting, measuring, analyzing, and interpreting data from social media platforms (like Facebook, X/Twitter, Instagram, LinkedIn, YouTube) to gain **business insights** and support **data-driven decisions**.

In simple terms: It helps organizations understand **what people are saying online**, how they **engage with content**, and how it impacts **business performance**.

### Purpose of Social Media Analytics

- Monitor **brand reputation** and customer sentiment.
- Measure **campaign performance** (likes, shares, reach, engagement).
- Identify **trends and customer preferences**.
- Track **competitors' strategies**.
- Improve **customer service and engagement**.
- Support **marketing and sales strategies**.

### Key Metrics in Social Media Analytics

1. **Engagement Metrics**
    - Likes, shares, comments, retweets, clicks.
2. **Reach & Impressions**
    - How many people saw the post.
3. **Conversion Metrics**
    - Leads generated, sales from campaigns.
4. **Sentiment Analysis**
    - Positive, negative, or neutral customer opinions.
5. **Follower Growth**
    - Increase/decrease in audience size.
6. **Virality**
    - Speed and scale of content sharing.

### Techniques Used in SMA

- **Text Mining & NLP** → Analyze customer reviews, tweets, posts.
- **Sentiment Analysis** → Classify emotions (positive/negative/neutral).
- **Trend Analysis** → Detect emerging topics and hashtags.
- **Network Analysis** → Study influencer impact and community structures.
- **Predictive Analytics** → Forecast future engagement or campaign results.
- **Visualization Tools** → Dashboards, heat maps, word clouds.

**Tools for Social Media Analytics**

- **Google Analytics** (for web + social integration)
- **Hootsuite, Buffer** (social media management & reporting)
- **Sprout Social** (analytics + engagement)
- **Brandwatch, Talkwalker** (advanced sentiment & trend analysis)
- **Socialbakers, Zoho Social** (campaign insights)

**Applications of Social Media Analytics**

- **Marketing** → Track campaign ROI and effectiveness.
- **Customer Service** → Respond quickly to complaints/queries.
- **Brand Management** → Monitor reputation and crises.
- **Product Development** → Gather customer feedback for innovation.
- **Competitor Analysis** → Benchmark performance against rivals.
- **Political Campaigns** → Measure public opinion and engagement.

**Advantages**

- Provides **real-time feedback** from customers.
- Helps organizations stay **customer-centric**.
- Identifies **trends and opportunities** quickly.
- Improves **marketing ROI**.

**Limitations**

- Data can be **noisy and unstructured**.
- Sentiment analysis may be inaccurate due to **sarcasm or slang**.
- Requires **constant monitoring**.
- Privacy and **ethical concerns** in data collection.

**Real-World Example**

Example: **Coca-Cola** uses social media analytics to:

- Track **brand mentions** globally.
- Identify **emerging consumer trends** (e.g., health-conscious drinks).
- Engage with customers via targeted campaigns.

This helps them **adapt strategies quickly** and maintain strong brand presence.

**In short:**
**Social Media Analytics (SMA)** helps businesses extract insights from online interactions to improve **brand image, marketing, customer engagement, and decision-making**. It combines **data mining, text analytics, and BI tools** to make sense of the huge volume of social media data.

# Lecture 31

**Behavioral Analytics**

**What is Behavioral Analytics?**

**Behavioral Analytics** is the process of **tracking, analyzing, and interpreting user actions and behaviors** (clicks, purchases, browsing, interactions) to understand **how people engage with a product, service, or system**.

In simple terms: It answers **"What do users do, and why do they do it?"**

**Purpose of Behavioral Analytics**

- Understand **user journeys** (how customers interact step by step).
- Identify **patterns and habits** in user behavior.
- Detect **bottlenecks** or drop-off points.
- Predict **future actions** based on past behavior.
- Improve **customer experience and engagement**.
- Support **personalization** and targeted marketing.

**Data Sources for Behavioral Analytics**

- **Websites & Apps** → clicks, page visits, time spent.
- **E-commerce** → purchase history, cart abandonment.
- **Social Media** → likes, shares, interactions.
- **IoT Devices** → usage frequency, preferences.
- **CRM Systems** → support tickets, communication logs.

**Techniques Used**

1. **Clickstream Analysis** → Tracks how users navigate websites/apps.
2. **Cohort Analysis** → Groups users based on common behavior/time period.
3. **Funnel Analysis** → Analyzes steps leading to conversion (e.g., product view → add to cart → checkout).
4. **Segmentation** → Divides users into groups (loyal, new, at-risk).
5. **Predictive Modeling** → Forecasts user actions (e.g., churn likelihood).
6. **Heatmaps** → Visualizes where users click or spend most time.

**Examples of Behavioral Analytics**

- **E-commerce** → Understanding why customers abandon carts.
- **Streaming services (Netflix, Spotify)** → Recommending movies/songs based on watch/listen history.
- **Gaming** → Analyzing player behavior to improve game design.
- **Banking** → Detecting fraud by monitoring unusual account behavior.
- **Healthcare** → Studying patient lifestyle habits for preventive care.

**Advantages**

- Improves **customer engagement & retention**.
- Provides **personalized recommendations**.
- Detects **risks** (fraud, churn).
- Enhances **user experience (UX)**.
- Increases **conversion rates**.

**Limitations**

- Requires **large volumes of behavioral data**.
- Privacy concerns (tracking user behavior).
- Complex analysis due to unstructured data.
- May need **AI/ML tools** for deeper insights.

**Real-World Use Case**

Example: **Amazon** uses behavioral analytics to:

- Track customer browsing & purchase history.
- Recommend products (collaborative filtering).
- Send personalized promotions.
- Detect fraud in unusual order patterns.

This drives **higher sales, customer loyalty, and security**.

**In short:**
**Behavioral Analytics = Understanding user actions.**
It helps organizations track **what users do, why they do it, and what they are likely to do next**, enabling **personalization, engagement, and better decision-making**.

# Lecture 32

**Iris Dataset**

**Introduction**

- The **Iris dataset** was introduced by **Ronald A. Fisher in 1936**.
- It is one of the **oldest and most widely used datasets** for testing classification algorithms.
- Contains **measurements of iris flowers** from three different species.

It is mainly used for **pattern recognition, classification, and clustering tasks**.

**Structure of the Dataset**

- **Number of Samples (rows):** 150
- **Number of Features (columns):** 4 numerical features + 1 categorical class label
- **Classes (Species):** 3 classes (50 samples each)
    1. **Iris-setosa**
    2. **Iris-versicolor**
    3. **Iris-virginica**

**Features (Attributes)**

Each flower sample has 4 measurements:

1. **Sepal Length (cm)**
2. **Sepal Width (cm)**
3. **Petal Length (cm)**
4. **Petal Width (cm)**
5. **Class (Label)** → Species of Iris flower

**Example Records**

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species |
|---|---|---|---|---|
| 5.1 | 3.5 | 1.4 | 0.2 | Iris-setosa |
| 7.0 | 3.2 | 4.7 | 1.4 | Iris-versicolor |
| 6.3 | 3.3 | 6.0 | 2.5 | Iris-virginica |

**Uses of the Iris Dataset**

- **Classification** → Train models to classify flowers into 3 species.
- **Clustering** → Group similar flowers without labels (unsupervised learning).
- **Data Visualization** → Plot petal vs sepal dimensions to see separation.
- **Dimensionality Reduction** → Apply PCA to reduce from 4D → 2D for visualization.
- **Educational Purposes** → Beginners use it to learn ML techniques.

**Why Iris Dataset is Important?**

- **Simple & Clean** → Small dataset, easy to understand.
- **Balanced** → Equal number of samples per class.
- **Benchmark** → Used to compare different ML algorithms.
- **Visualization-Friendly** → Easy to plot in 2D/3D for learning.

**In short:**
The **Iris dataset** is a **classic dataset in machine learning** containing measurements of 150 iris flowers across 3 species, used for **classification, clustering, and visualization tasks**.